

CHAPTER 4

TACTICS



PHOTO: END REVIK

Stian Christophersen sending Vingsand Pirates (Font 8a), Vingsand, Norway.

TACTICS ARE IMPORTANT for performing well on a route or boulder problem, but many of the tactical elements get overlooked. The most common is to focus on how to climb the route – like how to solve the different sequences, where to clip from and where to rest. You can also be tactical in how you prepare mentally before climbing and lay down a plan for how to think during the climb. In a broader sense, tactics are strategies for when to climb, who to climb with and how many attempts to have. What follows is an overview of the different tactical elements and why they are so crucial for performance. We then look at specific examples of tactics for on-sighting or redpointing routes and boulder problems, and tactics for competition climbing.



PHOTO: ALEX MANELIS

Maria Davies Sandbu enjoying a warm sunset wall in Bishop, USA.

‘TO CLIMB DURING PRIME CONDITIONS, YOU NEED TO KNOW AT WHAT TIME DURING THE DAY THE SUN IS ON THE WALL, AND WHAT SUITS YOU THE BEST.’

WHEN SHOULD I CLIMB?

To perform at your best, it's important to be aware of when the conditions are prime for doing just that. If you're planning a training session where you're going to try your indoor project, you should plan it for a time where you have the necessary mental and physical surplus. Are there days where your agenda is less full than others? Rushing from a hectic day of work and family life to squeeze in a short training session before moving on to other activities is seldom synonymous with being fully rested and having complete focus on performance. Choose days where you have more time and more energy, and times where your route is less likely to be busy.

With outdoor projects, you also must consider the weather and conditions. In general, you want to find the optimal balance between it being cold enough for maximum friction, but warm enough for your fingers not to go completely numb. Vertical walls, on rocks such as gritstone, granite and sandstone, are more friction dependent because of the fewer and smaller features on the wall, and therefore require lower temperatures than steeper limestone climbs which will often involve larger and better holds.

To climb during prime conditions, you need to know at what time during the day the sun is on the wall, and what suits you the best. Before going out, check the weather forecast and consider if it's supposed to be sunny or overcast, what the air humidity will be and if there's going to be any wind. By only going out on the good days, and being warmed up and ready to pull hard when the conditions are prime, you'll save both skin and energy by not wasting attempts during bad conditions. On a bigger scale, you should also consider at what time of the year to climb. It is usually better to try friction-dependent boulder problems and routes during the winter, or cooler days in spring and autumn, rather than in the middle of summer. The summer is often too hot for hard ascents unless it's really windy or the holds are relatively large. Remember that how much our hands sweat and how well we cope with climbing in the cold differs for everyone.



PHOTO: MIKE HUTTON

Pete Whittaker searching for friction on Stanage without Oxygen (E5 6c/Font 7a+), Stanage Edge, Peak District, England.

FRICTION

Friction is important to us climbers and is in many cases crucial to whether an attempt is successful or not. You might have experienced the difference between holding a sloper outdoors on a cold day and on a hot day. On a hot day, the hold can seem impossible to hang on to, while in cooler conditions it can feel like a pretty decent hold. Some holds have a surface that seems to always provide good friction, while others are smooth and slippery – even though they might look entirely the same. Footholds, and especially smears, can also be perceived as better when the temps are lower, and you won't slip off as easily if the footholds have a rough surface texture. So what is friction? And how can it be utilised to its full potential?

Friction is a force acting between two surfaces that are touching. To get a better understanding of how friction works, picture the surfaces as being uneven and all their tiny divots and protrusions catching on each other, hindering the surfaces from sliding. If a surface is rough, there will be more of these divots and protrusions, and the friction will be better. And likewise, the friction will also be better if the contact surface is large because a larger contact surface means more divots and protrusions. The material properties of these two surfaces can admittedly change as a result of different temperatures. This is largely a result of the materials becoming softer as they get warmer, which affects the friction.

ROCK TYPES

Rocks with round crystals have less friction than rocks with sharp crystals, and rocks with more crystals have better friction than rocks that have fewer. Plastic climbing hold manufacturers have begun to exploit this trait. Some brands offer the same hold with either good or poor friction. The difference in friction is caused both by the manufacturer varying between a sharper or more rounded texture, and varying the amount and size of the texturising additive. By climbing on holds with varying degrees of friction you'll get better at adapting how much grip force it is necessary for you to exert. If the friction is good, you'll be able to relax more, but if the friction is bad, you'll have to grip harder. Using just the right amount of power when climbing is one of the most important techniques to master. Climbing on holds with varying degrees of friction is one of the better ways to train this technique.

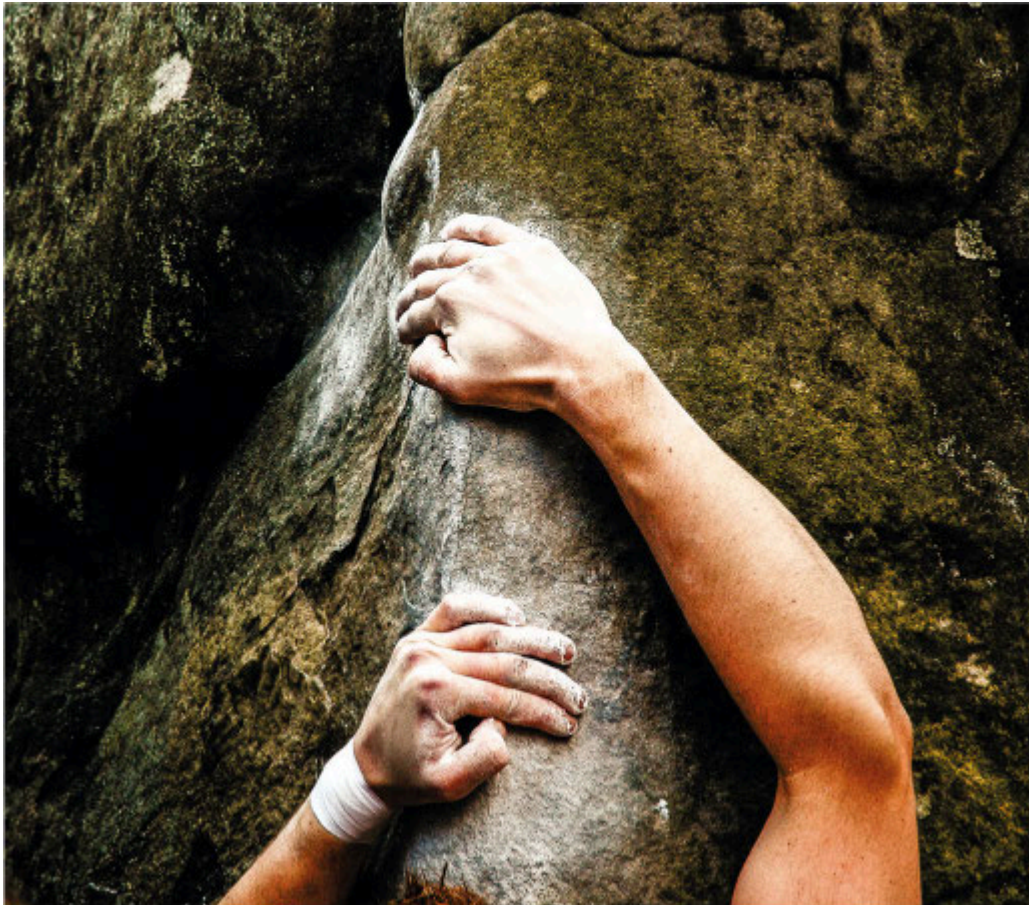


PHOTO: ANDREAS EIDE

HANDHOLDS

You have probably noticed how some handholds get really slippery in the heat. This is mainly caused by your fingers sweating more. But you may also have experienced sliding off holds because it was too cold. It might feel

like your skin is like glass trying to grip holds made of glass. Friction works because your skin moulds to the imperfections in the rock. This requires strong and flexible skin, and the ideal temperature is between 0° and 15° Celsius. If it's warmer, your skin will become too soft because of sweat. Your skin will catch on the protrusions of the rock, helping you grip, but it'll give and tear, and you'll slide off the hold. After a while, your skin wears down and you'll sweat even more and thereby slide even more. If, on the other hand, it's too cold, your skin will become hard and it won't be elastic enough to adapt to the texture of the hold, and again you'll end up sliding off.

Because friction is dependent on the contact surface, try to always maximise this. Let your fingers follow the shape of the hold, getting as much skin as possible in contact with the grip surface. By doing this you'll also end up not wearing down your skin as fast. A smaller contact surface will increase the pressure against the skin that is in contact with the hold. Your skin can only take so much pressure before it tears, so this is quite risky. You've probably noticed this effect when crimping small, sharp holds outdoors. Your skin wears down faster and tears.



•Most people use chalk to absorb sweat and keep their skin dry and not too soft. But too much chalk will only make things worse. It's been proven that dry, chalkless hands yield better friction than dry chalked hands. Sweaty fingers are worse though, so chalk sparingly and as needed!

•Brush the holds! The small indentations in the hold which your skin needs for friction pack up with chalk and dead skin, ruining the friction.

•Choose cooler days for sending your outdoor project. A cool breeze can also help!

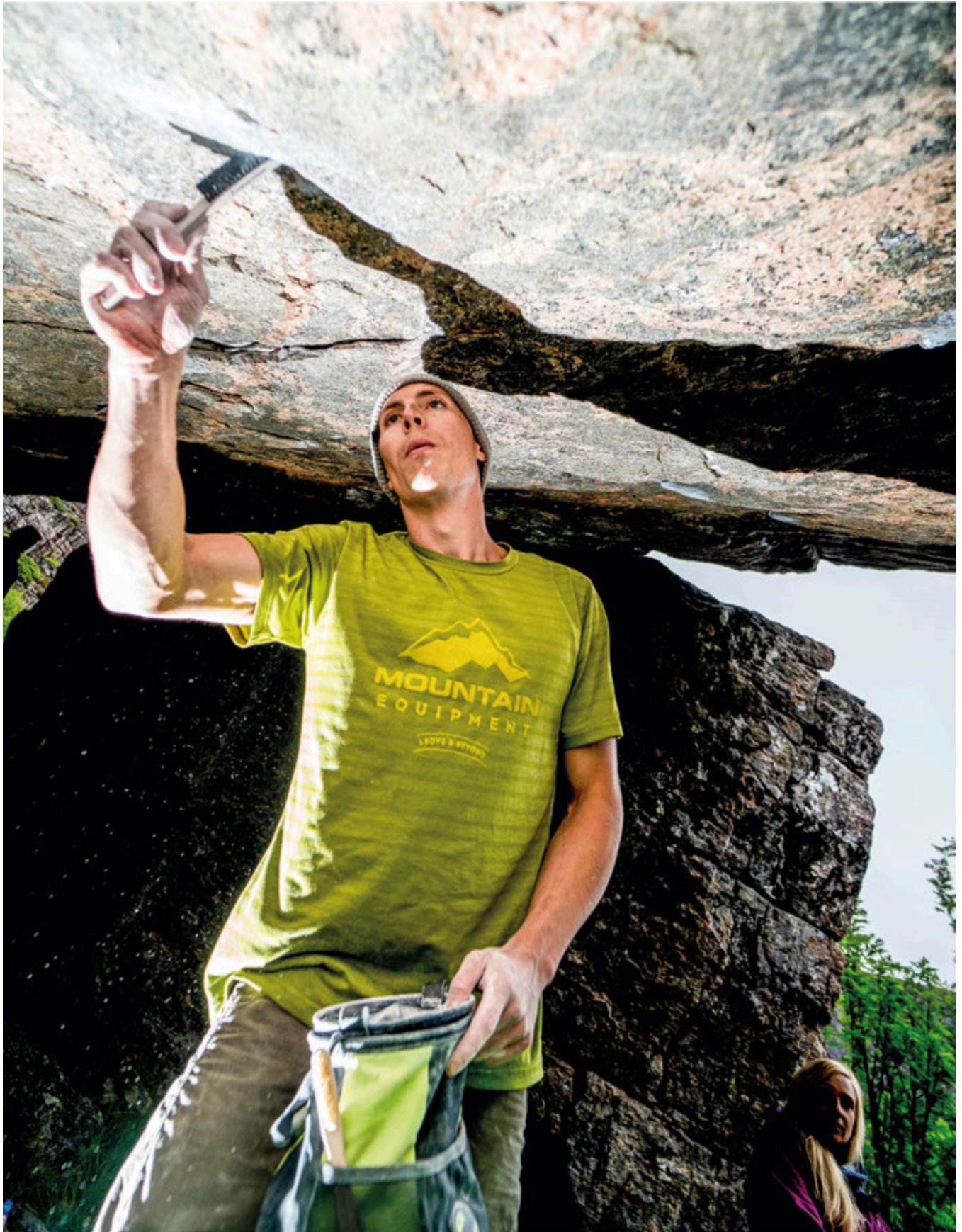


PHOTO: BJØRN HELGE RØNNING

Brushing the holds free from chalk and dirt is essential for optimising friction.



PHOTO: ENDRE VIK

Maria Davies Sandbu sending Flaskeposten (Font 7b+), Vingsand, Norway.



FOOTHOLDS

The rubber in your climbing shoes is designed to perform at its best when the temperature is between 0° and 15° Celsius. Within this interval, the rubber is flexible, yet adequately firm. At lower temperatures the rubber won't adequately mould to the texture of the foothold, and at higher temperatures it will easily deform – both of which increase the risk of your foot slipping.

We have highlighted the importance of lowering your heel to maximise friction when smearing. This is because it increases the contact surface between your shoe and the hold, so there are more features for the rubber to catch on. If the contact surface is too small the pressure will be very high and the rubber will roll and your foot will slip off.

Climbing shoes need to have both good edging properties and smearing ability. Harder rubber is better for edging, softer rubber is better for smearing. Climbing shoe manufacturers, therefore, have to compromise between the different properties climbers might demand from a shoe and its rubber.



•**Wipe your shoes before climbing! Dirty shoes add considerable wear and tear to holds, and reduce friction.**

•**Preheat your shoes if it's cold outside. You can do this by warming them inside your jacket, or by rubbing the sole with the palm of your hand to heat the rubber; this way you'll also remove loose dirt and debris from the sole.**

WHO SHOULD I CLIMB WITH?

Some climbers get along with just about anyone, while others need to surround themselves with a particular type of people in order to perform at their best. Some climb better when there are lots of people watching, while others tend to buckle under the pressure and prefer to climb in solitude. No matter what type of climber you are, we dare say you'll always perform better if you're climbing with people that make you happy and stoked. There's nothing wrong with being selective in choosing who to climb with when trying your project, and by knowing yourself well enough you can plan who to climb with. In addition to your partner getting you psyched to try hard, it's also important that you trust this person. You need to trust in their ability to belay you when you're lead climbing or know that they will move the pads and spot you if you're bouldering. If you're afraid of falling, it will be hard for you to focus on the climbing and not the falling.

It's often a good idea to have some thoughts on how time should be shared between you and the person you're climbing with. You need to know if your partner's project is close to yours, and how much time you will need for warm-ups and attempts. If you're trying different projects at different ends of the crag, you might need a plan for how much time to spend in each location, so that you know when to warm up and when to climb. Staying warm and ready to perform for a whole day is challenging, so the best solution might be to split the day in half, and start warming up for your own project after your partner has finished theirs.





PHOTO: TERJE AAMODT

Rannveig Aamodt after one attempt too many.

HOW MANY ATTEMPTS CAN I HAVE?

There's a big difference between trying a long and pumpy route and a short and technical route. The same goes for boulder problems: some are long and drain your energy every attempt, while others have just that one really hard crux move that needs to be practised to nail it down. Also crucial to how pumped you'll get is whether the crux is at the start or the beginning; this will affect how many attempts you can expect to have before getting too tired. These route and boulder problem characteristics should be in the back of your mind when planning your strategy for the day. The same goes for sharp holds. You might only manage a couple of attempts before your skin gets too soft or cracks, so you need to make the most of every attempt.

It is therefore important to form an opinion on how many good attempts you can have on a route or boulder problem. The number of attempts with a high probability of success will set the standard for how long you should rest between each attempt. Some routes and problems can have technically difficult moves that require many attempts to figure out, and in our experience, it's better to not rest for too long, lest you 'forget' how to do the move. Still, we cannot emphasise enough how important it is to take proper rests between attempts, and many of you are sinners in this regard – especially those of you who are pebble wrestlers! On routes and longer boulder problems, it may be appropriate to rest for up to one hour between attempts. This requires a good portion of patience and self-discipline, but as long as you know what to do and spend little effort and skin on finding better solutions, it's better to rest for too long than not rest enough between attempts. With long breaks, you need to keep your body temperature up between goes. It takes a lot of effort to do a complete warm-up multiple times, but with some warm clothes on and by staying active after your first go, it can suffice to only warm up your fingers and arms before pulling back on. A portable fingerboard can be an effective aid, and you'll be able to warm your fingers up without tearing your skin.

‘MARTIN USUALLY JUST SITS AND STARES AT HIS FINGERS IN CLASS.’

FROM MARTIN'S HIGH SCHOOL YEARBOOK

SKINCARE

Fingertip skin is important for climbers and can be the source of much frustration and worry. Good skin is important for optimal friction on the holds and to get the most out of a day of climbing. Unfortunately, skin wears fast, and if you're unlucky, it might even crack or tear. So, what can we do to maintain our skin, and what do we do if the skin just can't take it any more?

First and foremost we would like to point out that skin quality varies from climber to climber. Some people have hard and inelastic skin that doesn't sweat a lot, while others have softer and sweatier skin. Skin also becomes harder as it gets colder and softer when it's warmer. Hard skin is better for coarse crystals on real rock and for indoor plastic holds with lots of friction, but it is a disadvantage on smoother holds with less friction. You should get to know your own skin and how it reacts to temperature changes, as this will affect how you should go about treating it.

As a general rule, your skin needs moisture. The best way to keep your skin moisturised is to drink enough water and to use moisturiser as needed. If your skin is too dry it will grow more slowly and become less elastic. If you generally have dry fingers and relatively hard skin it's important that you use moisturiser, but this will be beneficial for most people anyway, especially if they have sore fingers. We prefer a moisturising cream that isn't wax-based, as the wax will remain on the outside of the skin without penetrating it. It can be a good idea to use moisturiser a few hours before climbing, and definitely afterwards. If your skin is soft and prone to sweating there are some creams that act as an antiperspirant which you could consider trying.

Always trim your nails before climbing if they are too long. It's also a good idea to wash your hands first, especially if you've eaten oily or greasy food or if your hands are dirty. Dirt and especially fat on your fingers will reduce the friction, you'll slide more off the holds and your skin will wear faster. When you're climbing it's important to pay attention to your skin. Consider filing down any irregularities and smoothing out rough edges. Do this with a bit of sandpaper or a file when resting between attempts. Filing your skin is especially important outdoors where the crystals tear your skin more easily than indoor plastic holds do. If you continue climbing after your skin tears, chances are that your skin is going to catch on some crystals which could rip off larger pieces of your skin. To avoid unnecessary wear and tear to your skin you should also let it rest in between attempts. Warm and sweaty skin means poorer friction and your skin will wear faster, so give it some time to cool down. Remember to also chalk your hands and to brush the chalk away from the holds in between goes. For every time a hold slips, you'll lose some of that precious skin. When you have finished climbing for the day, consider filing down any unevenness. Especially before having a rest day, it can be wise to file the skin down, so it can have time to grow while you rest.

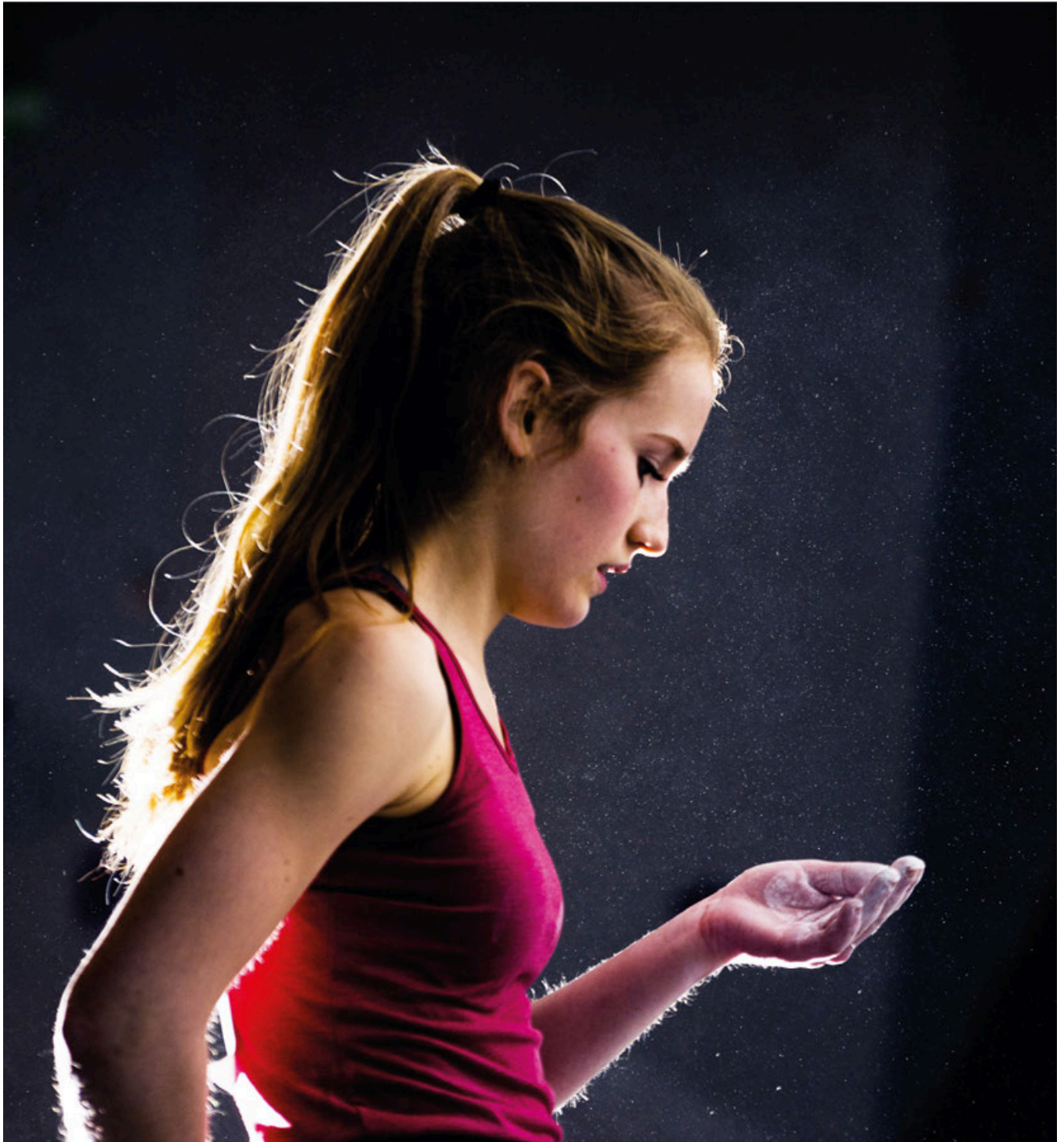


PHOTO: BJØRN HELGE RØNNING

Mia Støver Wollebøk assessing her skin before another attempt during the 2017 Norwegian National Bouldering Championships in Oslo.

If you've already climbed one move too far and your skin can't take it any more, you have two choices: you can either stop climbing for the day, often the wiser choice, or you can tape your finger and carry on. Below are three typical climber skin traumas and how to deal with them:

- A split is a crack across the finger, and you can get a split either at the joint or in between joints. Splits are caused by the skin being too hard and inelastic to cope with the pressure of pulling hard on a small hold. If you get a split you should file down the edges and let it rest. It can take several days for the skin to grow back to a point where you can start pulling hard moves without it splitting open again, especially if the split is near a joint, but it will take considerably longer if you don't file down the hard skin. If you have to keep on climbing you should tape your finger to stop it bleeding on the holds. Try also adding some moisturiser inside the split to soften the surrounding skin. This will reduce the chance of it splitting even further. Remember to cover the moisturised area with tape and remove any excess moisturiser from your other fingers.

- A flapper is when part of your skin gets torn away. This most often happens to the skin on the two innermost 'pillows' on your fingers and usually occurs on larger holds when you're pumped or when you're trying really dynamic moves. To avoid this, you should file down any hard calluses, as these will tear away more easily than softer skin. If you do get a flapper, cut the loose flapper off and tape your finger if you want to continue to climb. Since it might be painful to climb after getting a flapper, we recommend switching to less steep climbing with smaller holds that won't put so much pressure on the inner joints of your fingers.

- Thin skin is, as the name implies, when you've completely worn down the skin on your fingertips. Thin skin can be a consequence of having climbed on thin, sharp holds, and then wearing it down further on larger, slopey holds. To avoid it from happening, it's important to file the skin down as fast as possible if your skin starts to get small cuts and tears. It's important that the crystals don't catch and tear your already rough and torn skin any further. If you get thin skin on one finger, you most likely have thin skin on the other fingers as well, and it's time to take some rest and let the skin grow. Skin will often improve considerably with just one day of rest, though if it's very thin you might consider two days off. Thin skin is often a result of a lot of climbing, so perhaps a rest day is in order anyway. If, on the other hand, it's the last day of your climbing trip, you might want to just tape it up and go all in!



If you need to tape a fingertip, it will help the tape to stick better if you start with a small strip of tape along the finger first, before wrapping more tape around the finger.



Skincare – the essentials.

HOW DO I PREPARE MENTALLY?

Having a climbing project can sometimes feel like something of a curse. Every available moment at work or before bedtime is spent thinking about the different sections of a route. Uncertainty knocks at the door: is the route even possible? Do I have the best beta for the crux? How will I manage to clip the draws when I'm pumped? In the chapter on mental performance factors, we highlighted how regulating stress and being confident are important elements, and through good mental preparation we can build confidence and find the right stress level for performing.

First and foremost you need to build a foundation of underlying belief that you're capable of sending your project. By breaking a route or boulder problem down into smaller sections and getting confident that you can do each section, it will be easier to string each section together. Visualise every single move, every foot placement, every clip. Picture how it will feel to do the moves when you're pumped – and always picture success. There will be differences in what stress level is appropriate, depending on the route or boulder problem. If your project is a boulder problem consisting entirely of a few hard moves, you should be as psyched as possible to apply maximum effort on every attempt. If you're trying a long route, it won't be wise to go all-in at the first bolt, or you'll end up with no energy left for the top; a calmer approach and a lower stress level are required. The optimal stress level can also vary depending on the particular section of the route. Picture a technical top-out when bouldering or a tricky slab at the top of a steep route: following a few hard moves with maximum effort you must then reset, calm down, and climb slower so as to not make mistakes that will cost you the ascent. We have already covered how to train these mental factors, but there's a tactical element in choosing how to best utilise them.



If you can do the moves, you can do the route!

As part of the mental preparations, it's good to have some routines. These routines can be anything from practical elements like when to get up in the morning, when to eat and how to warm up, to purely mental exercises like inner dialogue, visualisation and regulating stress. Good routines help you to trust in your preparations and boost confidence and your faith in your own success.



PHOTO: ENDRE VIK

Stian Christophersen dispatches with Lynx (Font 8a), Vingsand, Norway.



PHOTO: MIKE HUTTON

Pete Whittaker on the second ascent of The Jasmine Corridor (E6 6c), Millstone Edge, Peak District, England.

HOW DO I CLIMB THE ROUTE AND THE DIFFERENT SECTIONS?

When you start the process of climbing a route or boulder problem, a good strategy can be to break it down into smaller sections. Figure out which sections are more difficult, where you can rest, and what climbing style and tempo you should apply to the different sections. By working through each individual section you can start to link them together into longer sections. Projects that might seem impossible at first, can be perceived to be far more doable if you work away at them, piece by piece.

Short and intense sections should be climbed as swiftly as possible, as your capacity for hard climbing over time is limited. Hard sections with bad feet and hard moves and body positions quickly drain your energy, so even though the total climbing time on a route can be quite long, it's important to climb these sections as quickly as possible and get to the next rest. This requires good tactical preparations. It's difficult to climb quickly and precisely without any mistakes if you haven't spent time finding the right beta and practising each section well. In addition, you need to memorise and visualise the sequences so that when you're climbing you're automatically both as technical and tactical as possible. Because you have to be fast through the hard sections and slow down for the easier ones, managing the change of pace is important. Use any available rests when you can, as they are places where you can calm down, both physically and mentally. Here you should try focusing on breathing deeply and calmly to recover as well as you can. Then you can climb with a higher tempo through the next hard section until the next rest. There will also be sections where you're forced to climb slower because the hand and foot placements need to be precise, placing high demands on balance and weight transfer. By splitting the route or boulder problem into sections where you set a predefined tempo, you're making it easier to put your tactics into practice and climb as energy efficiently as possible.

HOW TO REDPOINT A ROUTE

If you train specifically for a certain route, and you're willing to invest enough time into the project, it's very likely that you'll be able to climb it, but good tactics can drastically reduce the time it will take.

First and foremost, choose a route that motivates you to put in the effort needed. In many cases, a route is chosen because of its grade, how well it suits you and who's climbed it previously. There's nothing wrong with that, but we would challenge you to consider routes that appeal to you for other reasons. Perhaps there's a line that inspires you, a single move or section that sparks joy, or that the fact that the route actually doesn't suit you. A successful redpoint should mean something other than that you're capable of climbing a certain grade – it should also mean that you're developing as a climber and that you've had some good climbing experiences.

When you've chosen a route, figure out exactly what makes this route challenging. First, you need to figure out how to do the different moves and sections. Often you will be able to find a video or pictures of the route, but, better still, talk to others who have already either tried or climbed the route. It's a golden opportunity to discuss the beta and become better at problem-solving. A tip is to be conscious of your own climbing style and body, and don't give up if the standard beta isn't working for you! There's often more than one solution, and you should know your strengths and weaknesses in order to consider which solution will suit you the best. Try seeking beta from climbers that are roughly as tall as you and share your style of climbing. Yet there will always be individual differences regarding skill and preference, so even if you have a lot of good information, be open to finding your own solutions.

Many climbers spend too little time trying different solutions to different sections and end up failing on many attempts because they don't have the optimal beta.

There's a big difference between how the crux moves feel in isolation compared to when you've climbed into them from the ground up. Not just because you're tired, but also because you have built up a stress level that might not be optimal. We also see a lot of climbers failing or wasting unnecessary energy through the easier sections of routes. Nothing is as frustrating as falling off the easy last few moves because you didn't spend enough time figuring out the best solution. Despite moves and sections feeling manageable when you try them the first time, we recommend taking problem-solving seriously; try different alternatives before deciding which solution to go for.



PHOTO: STIAN CHRISTOPHERSEN

Martin Moberåten finishing a triple send of Høyspenning i Bissa (Font 8a+), Stange, Norway.

A TALE OF HOW CHANGING THE BETA ENDED IN TRIPLE SUCCESS

Høyspenningsprosjektet was a long-standing bouldering project at Stange in Norway. Stian had spent a few unsuccessful days on it a few years back, and a friend, Kenneth, started trying it in the spring of 2015. Martin joined in, and together they worked the project for a couple of days. Kenneth and Martin got close using two different betas, but Stian couldn't make either of the two solutions work. Frustrated, he started looking for other possible solutions, and by chance he found a foothold 10 centimetres further right than the one they had been using until then. Suddenly, the crux move was considerably easier, and after a short rest Kenneth, Stian and

Martin sent the problem first go with the new beta. Finding the balance between believing you have found the best beta and thus giving it all on every attempt, versus looking for new solutions, is notoriously difficult, but it's worth noting that often a small change is all it takes to make a big difference.

An important element to consider when you're working moves and sections is that you'll gradually start to tire, and your skin will soften and wear down. It might, therefore, be important to choose which sections you want to work, and then come down and have a rest before working the next section. Divide the route into sections and rank them according to difficulty. Work your way through the sections on the list one by one, starting with the most difficult. Remember to brush the chalk off the handholds and rubber off the footholds so that the friction is as good as possible.

In most cases, we choose routes that are possible within the current limits of our ability. The really big projects, on the other hand, may require you to specifically train certain qualities. You might be strong enough to do all the individual moves, but lack the endurance to link them all together. Or perhaps you need to get stronger on certain moves or types of handhold. This will require you to undertake specific training and re-evaluate the timeframe for when you can expect to climb the route. Furthermore, you need to evaluate if the route is conditions dependent, if the holds are sharp, and how the footholds are. This will help you to decide at what time of year and at what time of day is best for attempting the route, how many rest days you need to grow enough skin, which shoes to wear, and how many attempts you can have each day.

When you've found the right beta, you know what to train in order to succeed, you know what time of day you can expect the best conditions, how many attempts you can have, and how many rest days you'll need before climbing, that's when the mental preparations start. Visualise the moves so that you will remember them to the point. Picture yourself doing the crux moves, how you're going to climb before and after the crux, talk to yourself in a positive manner, and build confidence. Build up a positive expectation, look forward to going out climbing and treat being nervous as a positive thing. When you're visualising, you should feel your pulse rising and your palms sweating. Embrace all the emotions: really feel how you're not afraid of falling, that you can climb well even when pumped, and that you can still grip hard on every hold. Feel how you're applying weight on to the small footholds and convince yourself that your foot placements are rock solid.

When it's time to climb, it's important to have the right stress level and focus. Concentrate only on the tasks you've given yourself, and block out everything else. As Timothy Gallwey writes in his book *The Inner Game of Tennis*, performance equals potential minus distractions. To fully utilise your potential you must not allow yourself to be influenced by distracting thoughts or what is happening around you. Through simple tasks and a conscious focus, it will be easier to do exactly what you have planned to do. A simple way to reduce disturbances is to use a climbing partner you know well, to be prepared for the presence of other climbers at the crag, and to have a plan for what to do if the route you're trying is busy with other climbers. By being prepared, there will be less distractions, and if you're also able to block out negative thoughts, you'll have a much higher chance of success.



PHOTO: MIKE HUTTON

Naomi Buys on Green Lipped Mussels (F7c), Gordale Scar, North Yorkshire, England.



PHOTO: ANDY WICKSTROM

Maria Davies Sandbu climbing at Red River Gorge, USA.

An important part of the preparations is the warm-up. In addition to getting you physically prepared, the warm-up is a good opportunity to get in the right mood and at the right confidence level. You need to find your own warm-up routine, as what works best for others might not be the best for you, but it's common to stick to the same route or boulder problems every time or go through the same series of basic exercises and some fingerboarding. Many climbers also prefer to finish their warm-up on their project, trying individual moves or sections to refresh their memory and get a feel for the conditions.

All in all, we will fail more often than we will succeed, and being good at redpointing also means being good at dealing with failure. There might be days with prime conditions, when you're in great shape and your skin is good, and still you don't succeed. It is easy for the negative thoughts to come creeping in. Am I strong enough? Do I have the best beta? Should I just give up on the project and try something easier instead? There are no right or wrong answers to these questions, but if you want to succeed, you need to keep faith that you will. You also

have to continuously evaluate if you have the right beta. If you keep falling at the same move, you might want to consider whether your solution is optimal or not. If you do have the best beta for the crux, perhaps you can optimise the pump sections leading into the crux. It's challenging to keep balancing between believing in the plan you've made, and continuously considering other and perhaps better solutions – but you have to do this every time you fall. Likewise, perhaps you failed because you had put too much pressure on yourself and were too tense when climbing? Many times you might surprise yourself with an ascent when you least expected it. For example, after giving up, you might have tried one more time even though you were tired, because what the heck. When this happens, it's usually because you've put all expectations aside and you're more relaxed, just enjoying the climb. It's therefore important that you analyse the reasons for failing as objectively as possible so that you don't lose faith in your ability to succeed. By analysing what went wrong you can think of new solutions, visualise them, test them out on the route, and be ready for the next attempt.

In order to maintain motivation we also have to appreciate the process. If we go home sad and broken every time we don't succeed, we won't find much joy or learn anything from the redpoint process. As a rule of thumb, try to always find three positives for every negative, so that for every failure you analyse you will also find a sense of achievement and accomplishment. A positive attitude towards the process will make it more rewarding and create motivation every time you're on your project, even though not every time will yield the results you were hoping for.



PHOTO: CHRIS HOLTER/NORRØNA

A CLIMBER'S STORY

Norwegian climber Magnus Midtbø, born 1988, is a 17-time national champion in both lead and bouldering, and has had podium finishes in the World Cup and at the World Games. He's one of very few climbers worldwide to have on-sighted F8c+ and redpointed F9b.

MAGNUS VS NEANDERTHAL

Some routes are more memorable than others. Some of them leave scars that heal slowly – if at all. For me, Neanderthal in Santa Linya, Spain, is one of these routes.

Neanderthal is a long and steep route with relatively good holds and big moves. This style suits me well, and seeing as it was unrepeated after Chris Sharma made the first ascent in 2009, it appealed to me. And obviously also because of the grade, F9b, which made it one of the hardest in the world at the time. I started trying Neanderthal in 2012 and was progressing quickly. By the first week, I had solved all of the sequences and the crux, and soon I was able to do the whole route in two parts. You start by climbing a F7b+ up to a ledge where you can lay down and rest for as long as you like before the hard climbing starts right out of the gate after that. With the exception of one hold before the crux, there are no rests, with F9a/9a+ climbing up to the crux – a big dyno to a sloper. I could climb up to the crux, miss the crux move, pull back on and climb to the top after just the first few weeks of being there. I thought it wasn't going to take long before I sent it. Boy was I wrong.

After the first few weeks, there was no progression. The season in Santa Linya was coming to an end, and I had to leave for home empty-handed. Slightly disappointed, obviously, but believing I would succeed next time, I came back. And yet, as I returned the next season, history repeated itself. I was in better shape than ever, but still I couldn't do the crux move from the ground up. I would have training days in the cave where I would do multiple routes graded F8c to F9a, and I would do laps on some F9a routes, but still I couldn't do Neanderthal. After a while, the daily routine turned into a chore. I had one go every evening, and every time the same thing happened – I would fall on the same move without any sense of getting closer or further away than the day before. For the first time in my life, I hated climbing. I just wanted to finish the route and go home. Looking back now I realise this wasn't the right attitude to have for climbing at my limit, and I can honestly say that Neanderthal became more of a mental challenge than a physical or technical one.

'DESPITE THE ADVERSITY, I HAVE MANY FOND MEMORIES AND LEARNED A LOT THROUGH THE PROCESS.'

The uncertainty and frustration led me to overanalyse everything. Are the conditions perfect? Is my skin perfect? Am I well rested? If I didn't feel like everything was perfect I would lose all faith in doing any better than I had already done. And everything being perfect is an extraordinarily rare occurrence. Voltaire said that 'Perfection is the enemy of good', and perhaps I should have given that some thought – not everything had to be perfect for me to be able to send the route. In hindsight, I realise I should have climbed more in other places during the time I was projecting. I had some days at other crags where I felt stronger than ever before, and it was during this time I did my first F8c+ on-sight. I should also have tried other routes of the same grade which could have given me a sense of progression, instead of banging my head against the wall on the same move every day. I think that more variation in my climbing would have made my attitude more relaxed and lessened my obsession with finishing the route so I could finally move on with my life.

Despite the adversity, I have many fond memories and learned a lot through the process. For example, I've never gotten to know a route better than Neanderthal. Everybody who has ever had a climbing project is familiar with focusing on the details, but with Neanderthal I took it to the extreme. I knew every little recess of every hand and foothold, where to stop and breathe, and on which moves I needed to hold my breath to keep body tension. To experience how well you can know a route and the flow this gave my climbing is one of the fondest memories I

have. I also remember the life we had during the three seasons we spent down there. Life didn't consist of much more than eating, sleeping and climbing, and no matter how frustrating failing could be, I only did it for myself. Nobody in the whole, wide world could have made me come home before I wanted to, and it was enriching in many ways to fully focus on the one single target I had set myself. The adversity I faced was character building. Up until then, I had mostly been on a roll with my climbing. Even the hardest route I had done to date, Ali Hulk (F9b), took me 30 days, so there was always progress, and I have mostly accomplished the goals I have set myself. So, to first stagnate and then fail to realise something I initially thought was going to be quite doable, gave me new experiences which I later realised shaped me in a positive way.

Many people ask me if I would ever want to go back and finish the route. To this day it is still recognised as one of the hardest routes in the world. It would lift a heavy burden off my shoulders, but still, I have consistently said no. In the time after Neanderthal, I sought to embrace the joy of climbing and focused less on climbing hard routes and competitions because I grew tired of the focus on performance. I'm not sure I have the motivation required to shift my focus back to such a concrete performance goal as climbing Neanderthal. It will require a lot of training and a major shift in how I prioritise my life these days, but I also can't say that I'm not tempted. If I ever decide to try again, I will at least be fully committed to it, and even if I'm not at the same physical level as in the last rounds, I will at least be much better prepared mentally for the emotional rollercoaster such a project inevitably entails.

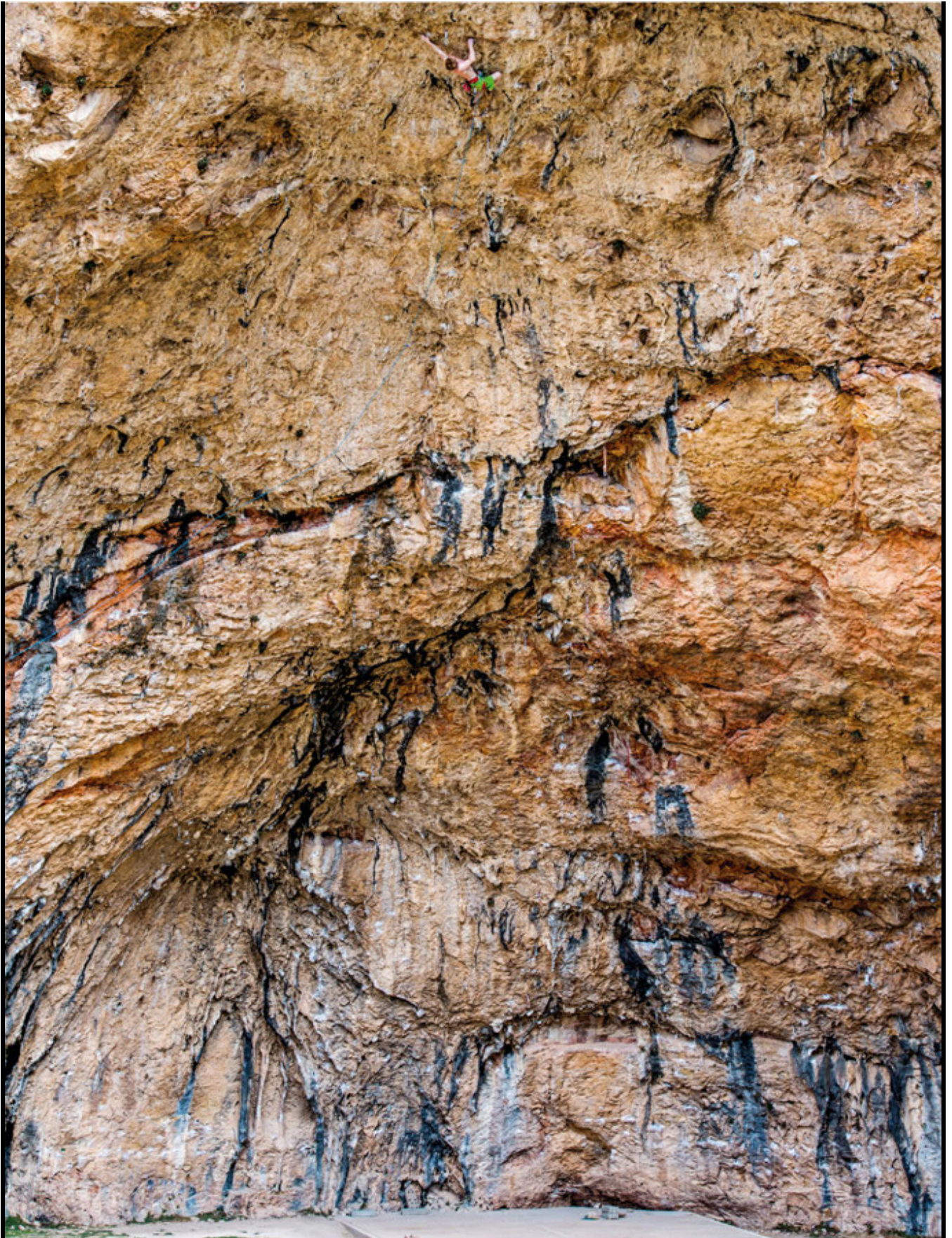


PHOTO: CHRIS HOLTER/NORRØNA



PHOTO: BJØRN SÆTNAN

Martin Mobråten climbing Wolverine (Font 8b+), Vingsand, Norway.

HOW TO REDPOINT A BOULDER PROBLEM

Boulder problems can be short or long, but in most cases, the moves are harder and more difficult compared to lead climbing. This means we must spend more time figuring out the beta and fine-tuning individual moves and short sections. While lead climbing is often about stringing together multiple sections in an effective way, bouldering is more about problem-solving a single move or one single section. This can be very frustrating but also very rewarding.

A boulder problem can go from being impossible for days, to suddenly being topped out in a relatively short amount of time, just because a missing piece of the beta puzzle falls into place. Since the individual moves are so important, conditions and skin play a more important role than when climbing routes, and redpointing a boulder problem often requires you to wait for prime conditions, when the chance of success is higher. If you don't have experience with this process yet, you'll be surprised at how a hold can go from totally useless to quite decent just because of lower temperatures and a cool breeze.

In the same way as when redpointing a route, you need to automate the moves and visualise the sequences. There's less margin per move than on a route, and there is no room for error, so good mental and tactical preparations are required. It is also important to regulate stress levels, and you need to decide which level is optimal.

For hard, physical boulder problems it's often advantageous to increase arousal and psyche yourself up so you'll be ready to really pull hard. Don't think about saving energy, go all in!

Adjusting to the right stress level is an art form you need to master in order to both pull as hard as you can, but at the same time be smooth and precise in your movement. You also need to account for the possibility of a delicate top-out requiring balance and precision following on from a steep and burly overhang; you will need to adjust your stress level accordingly.



Warming up on a fingerboard can be a good strategy for getting your fingers ready before getting to the boulders.

How you should warm up before climbing will also vary depending on what type of moves the boulder problem consists of. If the moves are technical and require coordination, part of the warm-up could be to practise individual moves to get the timing right before making proper attempts. If the individual moves are physically hard or the holds are sharp, it can better to warm up on other sections of the problem to optimise the beta, or on other boulder problems or a fingerboard if you're already confident on what to do when it's time to give it a go. Some problems require a high degree of flexibility, and in this case, your warm-up should include some mobility work similar to the moves you're about to do. It's important to take long enough rests in between attempts, even if they don't always seem necessary. On longer boulder problems where you fall because you're too pumped, it's obvious you need to take longer rests, but on shorter problems it might not always seem like you're too drained after an attempt. Yet you should still rest for long enough so that you're fully recovered for your next attempt, and so that your skin has a chance to cool down a bit to reduce sweating. If you're in a group of people all trying the same boulder problem, it's also important to let the holds rest after brushing off excess chalk and rubber, to lower the temperature and moisture level of the rock. It's difficult to say exactly how long you should rest for, and in our experience, it can be anywhere from ten minutes to one hour, where the latter requires you to warm your fingers up again for each new attempt. In general, longer rests are better than shorter rests when you know what to do and don't need to spend energy figuring out the beta. But still, muscle memory plays a role, and quite often the harder moves are also quite technical and require coordination. Taking long rests can therefore also be detrimental: you're well-rested, but aren't able to execute the move in exactly the same way as before your rest. One way to solve this problem is to rest for as long as you need, and then warm up again by trying that exact move or sequence to re-learn the timing of the movements.



STIAN VS EUROFIGHTER

Eurofighter at Røyken outside of Oslo isn't a fantastic problem. The crux holds are chipped, and you climb relatively low to the ground on the overhanging wall. Yet it's still the problem I've spent the longest working, and is my hardest ascent to date. What was it that grabbed my attention?

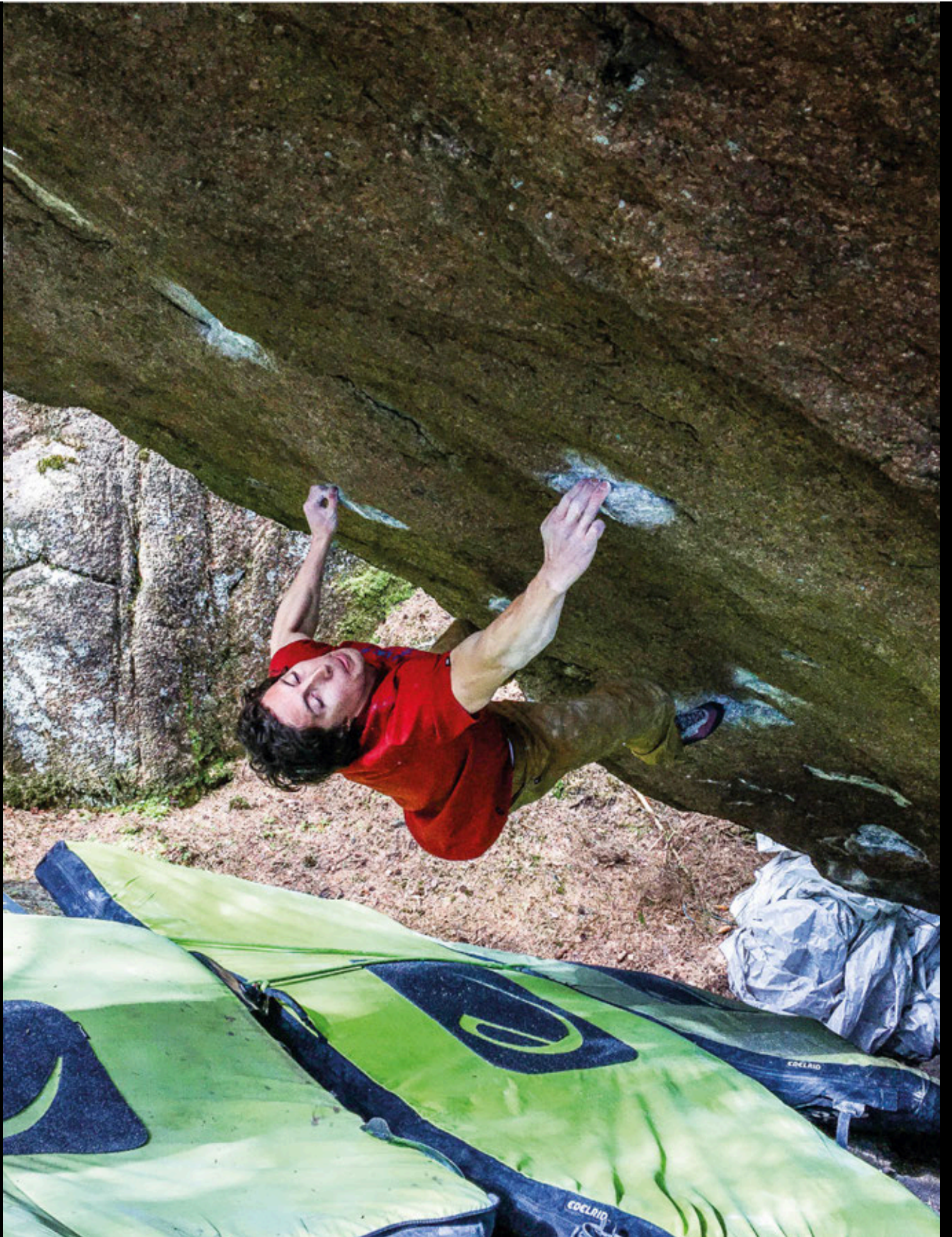


PHOTO: MARTIN MOBRÅTEN

Stian Christophersen sending Eurofighter (Font 8b+), Røyken, Norway.

‘ONLY THEN DID I REALISE WHAT WAS STOPPING ME FROM DOING THE CRUX MOVE, AND MY FOCUS SHIFTED.’

What was it that motivated me to spend several days of my life trying to climb and training specifically for this one boulder problem? First and foremost, it was hard. It was hard enough for me to have to improve, but not so hard that it felt impossible. Also, the moves are cool, which was perhaps the most important aspect.

The problem consists of eight moves on crimps through a very steep overhang, and it's both physically and technically demanding. The third move is the crux: a long deadpoint from a three-finger crimp with the right hand up to a three-finger crimp with the left hand. The body position is very scrunched with the left foot high, and the move itself has to be done quickly and with precision. Hitting the left-hand crimp was OK, but getting the timing right to stop the momentum proved to be difficult. I lost count of how many times I tried that move, or how many different betas I tried, and the first four days when I started trying it in 2012 were spent figuring out that one move. I think I did that move maybe one or two times during that period. The next few moves to the top aren't that hard, but the second to last move was really giving me hell. Even though I was able to do the move in isolation, doing it from just two to three moves below was something entirely different. Unlike the crux move this move isn't particularly technical, but it is a lot harder physically, and it felt like I had to get stronger in my arms, upper body and fingers if I was going to have any chance at linking it all together. Or get lucky.

When I started trying Eurofighter, there were others who were also psyched, and we had some good days underneath that roof. But as the days passed, there were fewer of us, and finally I ended up alone. Because the crux holds are chipped they're prone to seepage, and I often had to dry them off between every attempt. Being there alone, with nobody to help get me psyched, and constantly having to dry holds, my motivation started to dwindle. At the same time, a thought was starting to creep into my mind: even if I were to do the crux, I was probably going to fall off at the top anyway. 2012 ended with me packing up all the pads and leaving the boulder, thinking I would just have to come back and complete it some other time.

Years passed and I didn't return until 2016. During that period I had my second child and I was a lot more busy with work, but thanks to a highly motivated community I had trained well over several years. I had specifically focused on finger strength, something which I hadn't really trained during my two decades of climbing. Eurofighter had opened my eyes to weaknesses in my climbing that I wanted to address. The spring of 2016 was a surprisingly dry one, and there was hardly any seepage on the crux holds. The first day back on the boulder I realised that much had happened in the four years since I was last there. I had significantly more margin on all the moves, and I was quick to link the moves from the crux to the top. Day two I did the crux move several times, but fell four times at the second to last move, just as I had feared four years earlier. But this time my mindset was much more positive. I was convinced that I was strong enough and that it was just a matter of time.

On 15 May 2016 I went back with two of my best friends and climbing partners, Martin Moberg and Kenneth Elvegård. Conditions were good, I was well rested, I had slept well and my mind was calm. Most importantly, I was happy. During the last few weeks, I was having trouble sleeping, endlessly visualising the sequence. Everything was tuned, it was just a matter of doing what I had planned. After warming up and getting ready, I couldn't do the crux move. What had felt rather easy the last time, just wasn't happening. It took me about an hour before I realised I was too tense, and that I was focusing on the second to last move, and not focusing on the here and now and the crux. I was so ready to pull hard and to get the send that I couldn't

find the flow to get past the technical crux move. I was starting to think, 'I just want to get this done and over with'.

Only then did I realise what was stopping me from doing the crux move, and my focus shifted. The rest of the moves didn't matter – my only goal was to do the crux as many times as I could that session. If I didn't send it, the boulder wasn't going anywhere. I could just try it another day. After resting for half an hour, chatting about this and that with my good friends, I sat back down beneath the starting holds.



PHOTO: MARTIN MOBRÅTEN

Stian Christophersen on the first ascent of Eurofighter (Font 8b+), Røyken, Norway.

All I was thinking about was placing my left foot precisely and accelerating through my left hip. I trusted my body to remember the rest. The crux move was easy. The next two moves were easier still. After those two moves, you cut the feet and place your left foot high before bumping your left hand up into a sidepull. The second to last move relies on getting enough height in the sidepull. All that was in my head was to get the height and pull hard until everything went dark. The second to last move never felt easier than in that last attempt, and as I grabbed the lip of the boulder I became a bit perplexed at what had just happened.

Eurofighter isn't a hard boulder problem in a global context, but for me it was challenging in many ways. I had to get physically stronger, be conscious of who I brought with me, and I needed to find the balance between being relaxed enough to do the crux and determined enough to do the last hard moves. Two things separated 2012 and 2016: first, I had become stronger, and with it had become more confident in sending my project; second, I was more relaxed about whether I would end up sending it or not. In 2012 it was more important for

me to climb it, and I wasn't as interested in the process. Four years later I was more appreciative of the day-to-day progress and finding the joy in just trying it. It's a delicate balance, because if there is no will, there is no way. But it is apparent to me that I was better at that balancing act in 2016, and I realised how important it is to find motivation through mastery and not just results.



PHOTO: MIKE HUTTON

Jordan Buys on-sighting Primal Scream (E6 6a), Rathlin Wall, Fair Head, Northern Ireland.

HOW TO ON-SIGHT A ROUTE

Many of the tactical elements of redpointing are also important for on-sighting, such as at what time of day you can expect the best conditions, who you should climb with, and how to warm up. The big difference is that you don't know anything about the route apart from where the line goes and how hard it is. There's no information to be had from others regarding how to solve moves or sections, and you won't get to try any moves before the actual on-sight attempt. This requires you to picture different possible solutions based on what you can decipher from the ground. Following features, chalk marks and tick marks, you can often get an impression of which hand goes where, and where to place the feet. Many skilled climbers use binoculars to get a better look at the holds and spend a long time devising a plan for how to climb the different sections, and spot where they're likely to get a decent rest. When on-sighting it's more clear how to shift between different styles of concentration. When you find yourself at a rest you need to get an overview over the coming section. This will be the first time you get a proper look at the section, and you have to absorb and compute as much information as possible in a relatively short amount of time. A wide style of concentration is therefore crucial to the planning, but then you have to shift to a more narrow style of concentration and try to execute the moves as efficiently as possible.

Sometimes your plan will play out perfectly, but quite often you will stumble upon a surprise or two during the climb. Holds can be better or worse than anticipated, the body positions can be more challenging, and the rest you spotted from down below might turn out not to be a rest at all. You must therefore be open to using your intuition to solve the problems as they appear, and not completely lock yourself in to the initial plan. This requires a lot of experience and a large bank of movements. When we see experienced climbers on-sighting, it looks like they have already tried the route, because they execute each move with confidence and flow. Every move we do on the wall is unique, but there are many similarities, and if you're an experienced climber, it will be easier for you to recognise moves and solve new ones based on your experience.

The hardest part of on-sighting is climbing with confidence even when you're not sure if you have the right solution. Balancing between executing your plan and at the same time being open to new solutions is challenging and often leads you to become reserved, fearing you'll make a mistake and fall. It's important to build confidence in your ability to read the route, both while still on the ground and during the climb, and this confidence comes through experience. The warm-up is a good opportunity to build some of that confidence. First and foremost, stick to your regular warm-up routine, so that at least your preparations are familiar to you before you venture into the unknown. As a specific part of the warm-up, try climbing on other routes at the same crag or routes that have a similar style to the one you're going to on-sight. This can create the right stress level and will give you confidence in tackling the challenges to come, as well as familiarising yourself with the friction of the rock and the texture of the holds.

On-sighting a lot when you're at new crags or climbing walls will provide you with invaluable experience, but it's crucial that you go through your tactical and mental preparations as well as possible every time. Experience shows us that we fail far more often than we succeed when on-sighting, and so we must be proficient at analysing what went wrong and learn from our mistakes. Through trial and error we will gradually build experience and add to our movement bank, increasing our chance of success the next time.

HOW TO FLASH A ROUTE OR BOULDER PROBLEM

A flash differs from an on-sight in that you have information about how to do the moves, where the rests are, where the crux is and when the sun hits the wall. In short: you can and should collect any information that will help you succeed on your first attempt. The tactics you apply will, therefore, resemble both those for a redpoint and an on-sight; the main difference is finding confidence in the information you have.

Gathering beta is the key to a successful flash attempt. Not just figuring out the moves and sequences, but also anything regarding rests, clipping quickdraws, different climbing styles for different sections, and finding the right stress levels. You should consider who you're getting beta from. Choose climbers with roughly the same body size and shape, level, and climbing style as yourself, because most likely what works for them will also work for you. Then the mental preparations begin, with a particular focus on visualisation and stress regulation. By using visualisation techniques you can perfect the sequences in your head and then leave the climbing up to routine as your feet leave the ground. When flashing a route or boulder problem you'll know in advance where and what the character of the crux is, so you should decide whether to maintain a high level of arousal or calm down before climbing.

Practising making tactical decisions for redpoints, on-sights and flashes when training gives you valuable experience for those times when it really counts. It will at the same time increase the quality of your climbing every time you get on a new route: you'll get better at remembering moves, figuring out the beta, and building confidence in your ability to spot solutions from the ground and execute moves close to perfection first go.



PHOTO: MIKE HUTTON

Naomi Buys on Event Horizon (Font 6c+), Sampson's Stones, Lake District, England.

COMPETITION STRATEGIES

LEAD COMPETITIONS

Lead competitions consist of two qualification routes where you have one flash attempt per route. The route setters organise a demonstration of each route, and after that you can watch the other athletes ahead of you climb. Semi-finals and finals are on sight, where the athletes get to look at the route from the ground for six minutes before entering the isolation zone, from where the finalists emerge one by one without knowing how the previous climbers have performed. Whoever gets the highest up the route is crowned the winner.

In the qualification rounds it's important to know if the solutions presented are optimal for you. Differences in height, weight and climbing style can lead to different solutions, and even if it's likely that the solution presented by the route setters is the best, it is important to be open to other solutions. Take note of from which holds the quickdraws are clipped, and which sections seem the hardest. Memorise how the sections are climbed and visualise them. If possible, it can be useful to watch other climbers ahead of you to see if they are solving it the same way you were planning to, or if there are other options. Here it is important to choose climbers that you feel resemble your style, and not watch everybody. The best climbers usually get quite high on the qualification routes, but they might not be the best for you to watch if they are at a significantly higher level than you. Choose someone at roughly your level and see how they solve the route. This will give you a better idea of how your own attempt might look, and by limiting the number of other climbers you watch, you will have more time for your own preparations.

It's important to cooperate with others during the observation round before the final to get input on possible solutions. In the same way, as in the qualifiers, it's helpful to cooperate with just a few other climbers that you work well together with, and who you feel can provide valuable input. Look for footholds and positions to clip the draws from, in addition to the handhold sequences. This can give you valuable information on how to solve sections, and which sections are the most challenging. Decide what tempo to use for the different sections and where you plan to rest. There usually aren't many rests on a competition route, but sections requiring different strengths and techniques are all the more common, and this is a deciding factor when choosing the right climbing tempo. Once you have a clear mental image of the route, it can be a good idea to draw it on paper while you're waiting in the isolation zone to help you remember all the moves, holds and clipping positions. In the same way as when on-sighting outdoors, you have to believe in the plan you've made and climb confidently, but also be open to other solutions along the way.



PHOTO: RAGNHILD TRONVOLL

Tina Johnsen Hafsaas topping the final route at the 2016 Norwegian Lead Championships.

COMPETITION STRATEGIES

BOULDERING COMPETITIONS

International competitions consist of qualification and semi-final rounds with five and four problems respectively: five minutes per problem, with five minutes of rest between each problem. You have no information about the problem, and get points for topping a problem or for reaching the zone hold. The climbers are ranked based on their number of tops, number of zones, and finally number of attempts to top and number of attempts to zone. The finals consist of four problems where the climbers get four minutes per problem. Before the final round starts, the route setters explain where the first problem starts and stops, and where the zone hold is. The climbers then get two minutes to observe the problem, before moving on to the next. As with lead competitions, the climbers wait in an isolation area, and aren't able to watch each other climb and see how the others solve the problems. All climbers finish a problem before moving on to the next.

The only chance you get to collaborate with others in reading the problems is in the finals. As with lead competitions, it is a good idea to be conscious of who you choose to collaborate with. Both in terms of who you enjoy being with, and to get the best possible solutions that match your level and skill set. Solving problems on your own in an international competition format requires you to be able to read the problems and spot the possible solutions. A good tactic, therefore, is to practise by doing competition simulations and reading and solving problems on your own in a given timeframe. This way you will also learn to evaluate how many attempts you can have on each problem. Four minutes on a problem is not very long, and if it's physically demanding you might only have time for two good attempts and spend the rest of the time resting. With slabs and technical coordination moves, you might have time for a higher number of attempts, as these types of boulder problems won't drain your energy as much as the burly ones.

Controlling stress levels is one of the most demanding aspects of bouldering competitions. With five and four different boulder problems per round the route setters seek to challenge the climbers on different styles – some will be physically demanding, others technical and with tricky coordination moves. This requires different levels of arousal, and the route setters know how to take full advantage of this. The first problem will often be quite technical and require slow movements and precision. If you come out of isolation all fired up you will probably be too excited to climb well. You need to plan ahead and prepare for each problem, knowing whether to be calm and patient, or to explode, take risks and be on the offensive. It's important to stick to the plan, and it is also important to be able to reset between each boulder. It's easy to dig yourself into a mental hole if the first or even the two first problems don't go so well, but remember that you still have several chances left, so keep the positivity and confidence up. There can be many tops in a bouldering competition, but sometimes the final result comes down to a single top on the last problem. As a competitor, you won't be aware of the results, so it's important to stay strong and don't give up just because you couldn't top the first problems.



PHOTO: BJØRN HELGE RØNNING

Stian Christophersen competing at the 2017 National Bouldering Championships in Oslo.

TACTICAL PREPARATIONS

COMPETITION PLANS

For both forms of competition, it is important to have a competition plan. The purpose of the plan is to put you in the right mood before you perform, and when you do the same preparation every time it will be easier to recreate results. The plan should describe how your training will look up to the competition, an itinerary, and when and how to warm up. Detailed plans also include who you should be with and talking to, what focus you should have and a detailed warm-up routine.

DEALING WITH STRESS

Most people are nervous before competitions, and being nervous can be uncomfortable. You're restless, you're sweating and you have to go to the bathroom all the time. However, the increased stress should be seen as something positive. After all, you chose this yourself. You have trained for it and have prepared well. It means something to you whether it is going to go well or not, and it is a privilege to be in this situation. In order to perform, however, it is important to know how best to cope with a stressful competition situation. This is an important tactical element that greatly affects the outcome.

BINOCULARS, PEN AND PAPER

During lead competitions, binoculars can be worth their weight in gold when you need to take a closer look at the holds. Having more information about the holds can help you decide how to solve different sections and give you confidence about whether you can move dynamically to a hold or will have to do the move statically. It can also be helpful to make a sketch of the route after the observation period. This is good training for remembering moves and it's a helpful tool for visualisation of the route. This will increase your confidence, which will help alleviate some of the anxiety that characterises a competitive situation.



PHOTO: RAGNHILD TRONVOLL
